

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 4.3 Notes

Subspaces of R^n

Devotional Thought

A Major Disclaimer

We are skipping section 4.2 for now. Because of this, there are a few things you need to be aware of from a practical standpoint:

- Both the book and I will refer to *vector spaces*. We will use a special case definition (see next page) for now, but when only the MATH 201 class is meeting, I will start by providing the “full” definition.
- To accommodate this lack of knowledge, I am cherry-picking pieces of the sections we will cover up to our next test: 4.3 - 4.8, 6.1, 6.3 - 6.5, 7.1 - 7.3
- This also means you will need to be selective in what parts of the textbook you read between now and the second test. To help you better sift through content, I will present all of the relevant book examples.
- Such comments also refer to any outside resources you may be using, e.g. YouTube or other videos, Wikipedia, ChatGPT, Khan Academy, and the like. PLEASE DO NOT ACCIDENTALLY CONFUSE YOURSELF WITH EXCESSIVE, IRRELEVANT, AND/OR ADVANCED INFORMATION.
- The goal in all of this is to provide, with the limited time we have remaining in MATH 200, a simplified yet adequate understanding of certain fundamental topics in linear algebra: basis vectors, change of basis, linear transformations, eigenvalues and eigenvectors, and diagonalization.

Subspaces of R^n

For our immediate purposes (and until defined otherwise for the MATH 201 students):

A **vector space** is either R^n or a subspace of R^n .

Definition A nonempty subset W of R^n is called a **subspace** of R^n if and only if it is closed under both vector addition and scalar multiplication:

1. If $\mathbf{u}$ and $\mathbf{v}$ are in W , then $\mathbf{u} + \mathbf{v}$ is in W . Closure under vector addition
2. If $\mathbf{u}$ is in W and c is any scalar, then $c\mathbf{u}$ is in W . Closure under scalar multiplication

The main idea of a subspace is that it inherits the ten properties of R^n that were detailed in Theorem 4.2. This means that W acts as a sort of copy of R^k living inside of R^n where $k \leq n$.

To show that a set is a subspace of R^n we must check four criteria:

1. W is a subset of R^n (this is usually trivial).
2. The set W is nonempty, meaning something is actually in the set W .
3. For any vectors $\mathbf{u}$ and $\mathbf{v}$ in W , the vector $\mathbf{u} + \mathbf{v}$ is also in W .
4. For any scalar c and any vector in W , the vector $c\mathbf{u}$ is also in W .

Example [Example 1] Show that the set $W = \{(x_1, 0, x_3) : x_1 \text{ and } x_3 \text{ are real numbers}\}$ is a subspace in R^3 with the standard operations.

Example [Exercise 8] The set W of all vectors in R^2 whose first component is 2 is *not* a subspace of R^2 . Verify this by giving a specific example that violates the test for a vector space.

One standard way to show that W is nonempty is to check and see if the zero vector is in W ; this would need to be the case for a subspace to be a “copy” of some R^k .

In fact, the simplest subspace of R^n is the one consisting of only the zero vector, $W = \{\mathbf{0}\}$.

This subspace is the **zero subspace**. Another subspace of R^n is $W = R^n$ itself. Indeed, if n is an integer, then R^n will always contain these two **trivial subspaces**, and subspaces other than these are called **proper** (or **nontrivial**) **subspaces**.

Example [Example 2] Show that $W = \{(x_1, x_2) : x_1 \geq 0 \text{ and } x_2 \geq 0\}$ is *not* a subspace of R^2 .

Example [Exercise 2] Verify that the set

$$W = \{(x, y, 4x - 5y) : x \text{ and } y \text{ are real numbers}\}$$

is a subspace of R^3 .



**ME IN MATHS
CLASS USUALLY**



**ME WHEN THE
TEACHER STARTS TALKING
ABOUT SUBSPACES**

Subspaces of R^2

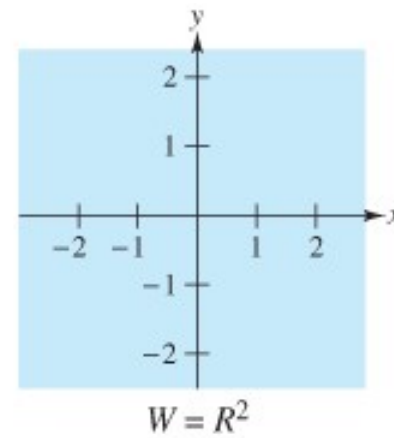
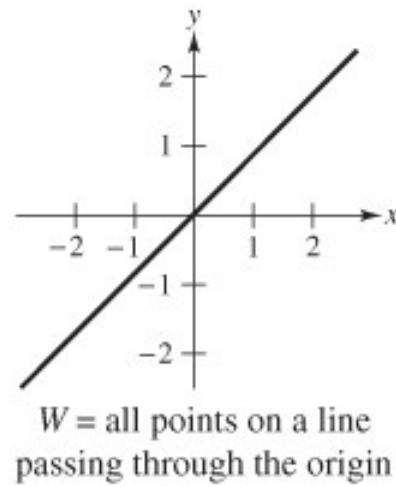
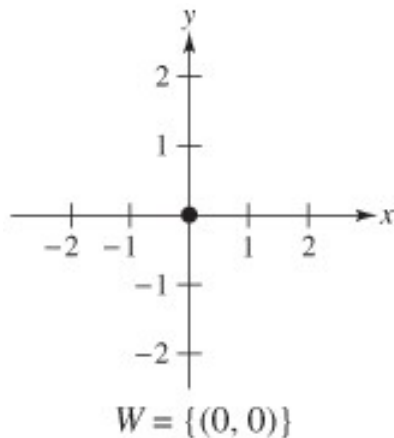
Example [Example 4] Determine whether each subset is a subspace of R^2 .

(a) The set of points on the line $x + 2y = 0$.

(b) The set of points on the line $x + 2y = 1$.

Of the two lines in the last example, the one that is a subspace of R^2 is the one that passes through the origin. This is characteristic of subspaces of R^2 . That is, if W is a subset of R^2 , then it is a subspace if and only if it has one of the forms listed below.

1. W consists of the *single point* $(0,0)$.
2. W consists of all points on a *line* that passes through the origin.
3. W consists of all of R^2 .



Example Show that the subset of R^2 consisting of all points on $x^2 + y^2 = 1$ is *not* a subspace of R^2 .

Subspaces of R^3

Example Determine whether each subset is a subspace of R^3 .

(a) $W = \{(x_1, x_2, 1) : x_1 \text{ and } x_2 \text{ are real numbers}\}$

(b) $W = \{(x_1, x_1 + x_3, x_3) : x_1 \text{ and } x_3 \text{ are real numbers}\}$

In the last example, note that the graph of each subset is a plane in R^3 , but the only subset that is a *subspace* is the one represented by a plane that passes through the origin. It can be shown that a subset W of R^3 is a subspace of R^3 if and only if it has one of the forms listed below.

1. W consists of the *single point* $(0,0,0)$.
2. W consists of all points on a *line* that passes through the origin.
3. W consists of all points in a *plane* that passes through the origin.
4. W consists of all of R^3 .

This tracks with our earlier observation about R^2 and can be thought of as saying that any subspace of R^3 is either a copy of $R^0 = \{\mathbf{0}\}$ (the single point), a copy of $R^1 = R$ (a line), a copy of R^2 (a plane), or R^3 itself.

Going further one could extend this intuition to R^n :

A subset of R^n is a subspace of R^n if and only if it is isomorphic to R^k for some nonnegative integer $k \leq n$.

Here *isomorphic* is the strict mathematical term for “is a copy of.” Intuitively speaking, isomorphic vector spaces have the same structure brought about by having the same dimension...

...but to understand what *dimension* really means, we need to discuss what is meant by a *basis* for a vector space...